Diabetes Classification

Understanding the Dataset - a self-contained guide to meanings, values and variable types

1. What is this dataset?

The supplied file contains 253,680 survey responses and 22 columns: 21 possible input features (X) and one target (y). The data come from health-related survey variables. Each row represents one survey respondent.

Source context: the assignment describes these variables as coming from the CDC Behavioral Risk Factor Surveillance System (BRFSS), a health-related telephone survey covering health-risk behaviours, chronic health conditions and use of preventive services. Variables may come directly from survey questions or from variables calculated from participant responses.

Why diabetes prediction matters: diabetes is a chronic condition involving impaired regulation of blood glucose. The assignment notes that early identification can support lifestyle changes and treatment, which motivates studying whether survey information can help predict diabetes risk.

Target: Diabetes_binary. In the supplied binary-classification CSV, 0 = no diabetes and 1 = prediabetes or diabetes. This is a binary classification problem. The older assignment handout also contains a line listing 0 = no diabetes, 1 = prediabetes and 2 = diabetes; that description belongs to a three-level coding and conflicts with the binary file description. For this course, use the values actually present in the supplied binary CSV: 0 and 1.

The classes are not balanced: 218,334 rows are class 0 and 35,346 rows are class 1 (13.9% class 1). There are no missing values in the supplied CSV.

2. How to use this guide

Do not decide how to analyse or preprocess a feature only from the number stored in the CSV. First ask what the value means. A value can be a measured amount, a count, a yes/no category, or an ordered category code.

Example	What the number means	Type
BMI = 30	A measured BMI value	Numerical measurement
PhysHlth = 10	10 days of poor physical health in the past 30 days	Numerical - discrete/count
HighBP = 1	Yes: respondent has high blood pressure	Categorical - binary
GenHlth = 4	Fair general health; 1-5 has an order	Categorical - ordinal
Age = 9	Age group 60-64; 9 is a category code, not age 9	Categorical - ordinal

3. Complete feature guide

Feature	Plain-English meaning	Values / categories	How to read the values	Variable type
Diabetes_binary	Target: diabetes status	0 = no diabetes; 1 = prediabetes or diabetes	Two outcome classes	Binary target y
HighBP	Whether respondent has high blood pressure	0 = no; 1 = yes	Yes/no category	Categorical - binary
HighChol	Whether respondent has high cholesterol	0 = no; 1 = yes	Yes/no category	Categorical - binary
CholCheck	Cholesterol checked within past 5 years	0 = no; 1 = yes	Yes/no category	Categorical - binary
BMI	Body Mass Index	12-98 in supplied CSV	Measured numerical value	Numerical measurement
Smoker	Whether respondent has smoked at least 100 cigarettes in entire life	0 = no; 1 = yes	100 cigarettes is approximately 5 packs; yes/no category	Categorical - binary
Stroke	Ever told that respondent had a stroke	0 = no; 1 = yes	Yes/no category	Categorical - binary
HeartDiseaseorAttack	Whether respondent has coronary heart disease (CHD) or myocardial infarction (MI)	0 = no; 1 = yes	Yes/no category	Categorical - binary

Feature	Plain-English meaning	Values / categories	How to read the values	Variable type
PhysActivity	Physical activity in past 30 days, excluding activity done as part of a job	0 = no; 1 = yes	Yes/no category	Categorical - binary
Fruits	Consumes fruit one or more times per day	0 = no; 1 = yes	Yes/no category	Categorical - binary
Veggies	Consumes vegetables one or more times per day	0 = no; 1 = yes	Yes/no category	Categorical - binary
HvyAlcoholConsump	Whether respondent meets the dataset definition of heavy alcohol consumption	0 = no; 1 = yes	Assignment definition: adult men >=14 drinks/week or adult women >=7 drinks/week	Categorical - binary
AnyHealthcare	Whether respondent has any kind of health-care coverage	0 = no; 1 = yes	Includes health insurance or prepaid plans such as an HMO; yes/no category	Categorical - binary
NoDocbcCost	Whether respondent needed to see a doctor in past 12 months but could not because of cost	0 = no; 1 = yes	Yes/no category	Categorical - binary
GenHlth	Self-rated general health	1 = excellent; 2 = very good; 3 = good; 4 = fair; 5 = poor	Ordered health categories	Categorical - ordinal
MentHlth	Days mental health was not good in past 30 days	0-30 days in supplied CSV	A genuine number of days	Numerical - discrete/count
PhysHlth	Days physical health was not good in past 30 days	0-30 days in supplied CSV	A genuine number of days	Numerical - discrete/count
DiffWalk	Serious difficulty walking or climbing stairs	0 = no; 1 = yes	Yes/no category	Categorical - binary
Sex	Respondent's sex	0 = female; 1 = male	Two category states	Categorical - binary
Age	Age group	1=18-24; 2=25-29; 3=30-34; 4=35-39; 5=40-44; 6=45-49; 7=50-54; 8=55-59; 9=60-64; 10=65-69; 11=70-74; 12=75-79; 13=80+	Ordered age groups; code is not age in years	Categorical - ordinal
Education	Highest education category	1=no school/kindergarten; 2=grades 1-8; 3=grades 9-11; 4=high-school graduate; 5=some college/technical school; 6=college/technical-school graduate	Ordered education categories	Categorical - ordinal
Income	Household income category	1=< $10k; 2=$10k-<15k; 3=$15k-<20k; 4=$20k-<25k; 5=$25k-<35k; 6=$35k-<50k; 7=$50k-<75k; 8=$75k+	Ordered income groups; code is not income in dollars	Categorical - ordinal

4. Detailed coding notes for ordered variables

Age is a 13-level ordered age-group variable, not age in years: 1=18-24, 2=25-29, 3=30-34, 4=35-39, 5=40-44, 6=45-49, 7=50-54, 8=55-59, 9=60-64, 10=65-69, 11=70-74, 12=75-79, 13=80+.

Education is ordered from 1 to 6: 1=no school/kindergarten, 2=grades 1-8, 3=grades 9-11, 4=high-school graduate, 5=some college/technical school, 6=college/technical-school graduate.

Income is an ordered household-income category: 1=< $10,000; 2=$10,000-<15,000; 3=$15,000-<20,000; 4=$20,000-<25,000; 5=$25,000-<35,000; 6=$35,000-<50,000; 7=$50,000-<75,000; 8=$75,000+.

GenHlth is ordered: 1=excellent, 2=very good, 3=good, 4=fair, 5=poor.

5. Important points before EDA

Duplicate rows: You may find rows that have identical values across all columns. Keep these rows in this assignment; do not remove them using drop_duplicates(). The dataset does not contain a respondent identifier that would allow us to determine that identical rows represent the same person recorded more than once. Because many variables have only a small number of possible values, different respondents can have exactly the same combination of recorded values. Therefore, use the supplied dataset without removing these rows.

- All columns are stored as numbers, but they do not all represent numerical measurements.
- Binary 0/1 features are categories with two states. Their values already provide a simple 0/1 representation.
- Age, Education, Income and GenHlth are ordered categories. The codes represent ordered groups, not measured distances.
- BMI is a numerical measurement. MentHlth and PhysHlth are numerical counts of days.
- There are no NaNs in the supplied CSV, so missing-value imputation is not needed. You still need to reason about feature meaning and representation.
- The target combines prediabetes and diabetes into class 1. Therefore the model predicts the supplied binary outcome; it does not distinguish prediabetes from diabetes.

Representation reminder for Logistic Regression: binary 0/1 variables are already encoded and need no additional categorical encoding; BMI, MentHlth and PhysHlth are genuine numerical values; Age, Education, Income and GenHlth are ordered categories and should be treated as ordered rather than as measured amounts.

6. EDA: ask a question before making a plot

Question	Useful evidence	What you may conclude
How imbalanced is the target?	Counts and proportions of Diabetes_binary	How common each class is in this dataset.
Does BMI differ between the two classes?	Class-wise distributions / boxplots	Whether the observed BMI distributions differ; not that BMI causes diabetes.
Does diabetes proportion differ between HighBP = 0 and HighBP = 1?	Crosstab with row proportions	Whether HighBP is associated with the target in this dataset.
Does diabetes proportion change across Age groups?	Proportional bar plot / crosstab	Whether the target proportion changes across ordered age groups.
Are two numerical variables related?	Scatterplot/correlation where appropriate	Association, not causation and not automatically model importance.

Memory rule: EDA describes patterns in the observed data. It does not by itself establish causation.

For categorical features, do not force ordinary numerical correlation onto category codes. Compare target proportions across categories using a crosstab or proportional bar chart. For genuinely numerical variables, scatterplots/correlation may be appropriate. In both cases, describe association rather than causation.

7. Classification questions that matter in this dataset

- Because class 1 is much less common, accuracy alone can hide poor detection of class 1.
- Recall asks: of the respondents who are actually class 1, how many did the model find?
- Precision asks: of the respondents predicted as class 1, how many are actually class 1?

Specificity asks: of the respondents who are actually class 0, how many did the model correctly predict as class 0?

Accuracy asks: of all respondents, what proportion did the model classify correctly?

- A false negative is an actual class-1 respondent predicted as class 0. A false positive is an actual class-0 respondent predicted as class 1.

Metric formulas: Accuracy = (TP + TN) / (TP + TN + FP + FN); Precision = TP / (TP + FP); Recall = TP / (TP + FN); Specificity = TN / (TN + FP).

- A probability is not yet a class. The threshold converts a predicted probability into class 0 or class 1.
- Changing the threshold changes the predicted classes and the FP/FN trade-off; it does not retrain the fitted model.

8. TRAIN and TEST

Split before learning from the data. For this classification problem, use the target classes when stratifying the split so that TRAIN and TEST contain approximately the same proportion of class 0 and class 1. In the code, this is done using stratify=y. Learn any data-dependent preprocessing and model parameters from TRAIN only. Keep TEST unchanged for final evaluation. Do not balance or resample the TEST set: it should remain an unseen evaluation set.

Core workflow: UNDERSTAND -> EDA -> SPLIT -> PREPARE -> FIT -> PREDICT -> EVALUATE -> INTERPRET